Game 1 Catch the Butterfly

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Build the game

Choose a butterfly sprite and a garden backdrop. Remove the cat if it is not needed.

Add the two scripts below to the butterfly. The first script starts the game; the second runs when the player clicks the butterfly.

```
when green flag clicked
show
go to random position

when this sprite clicked
play sound [pop] until done
go to random position
```

Choose a short sound available in your project. If the sprite is too small to click comfortably, increase its size.

Test and personalise

Click the green flag, then click the butterfly several times. Each click should play a sound and move it. Test a new game with the green flag.

Add a short instruction with a say block at the start. Check that the speech does not hide the butterfly for too long. Choose a different backdrop or sound.

Optional teacher-supported score: create a Score variable for all sprites. Set Score to 0 in the green-flag script and change Score by 1 in the click script.

Find and fix

If a click does nothing, check that the script belongs to the butterfly and starts with when this sprite clicked. If it disappears, add show at the start.

Explain

What starts each script? Show one improvement you tested.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

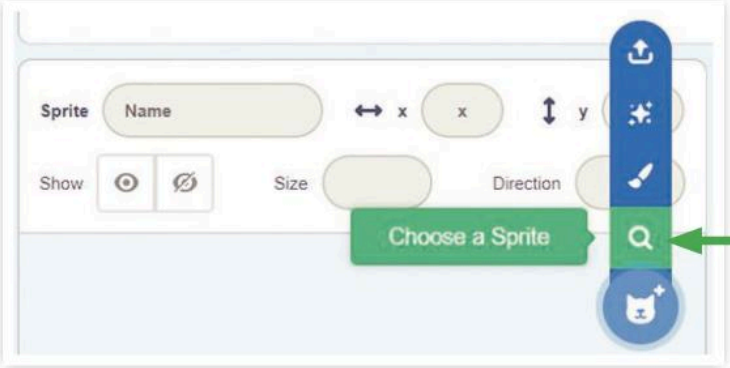
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

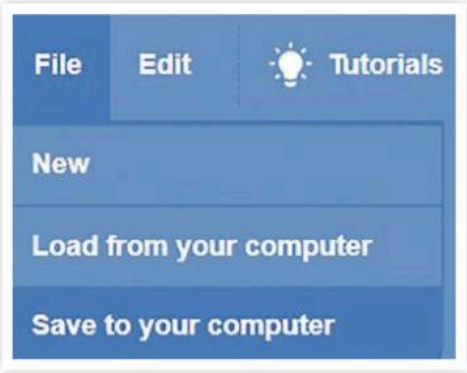
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.

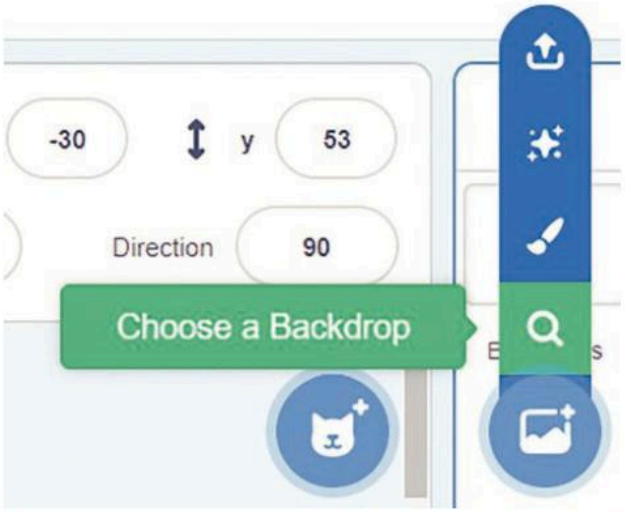


Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.



4.3 A start event

In this lesson

You will learn:

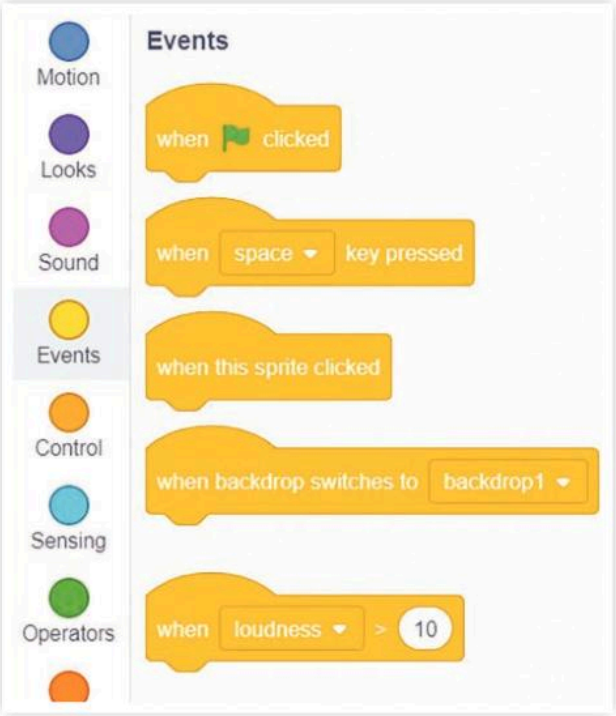
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

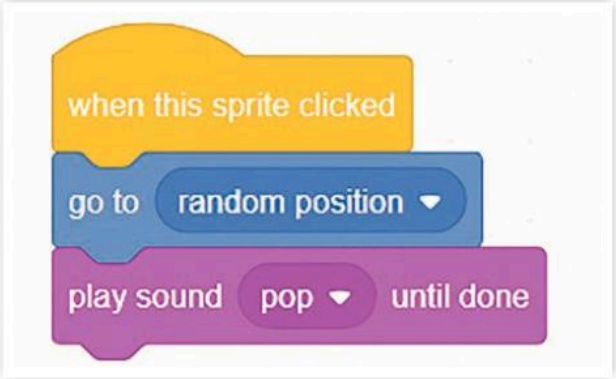


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.

4.6 Find and fix

In this lesson

You will learn:

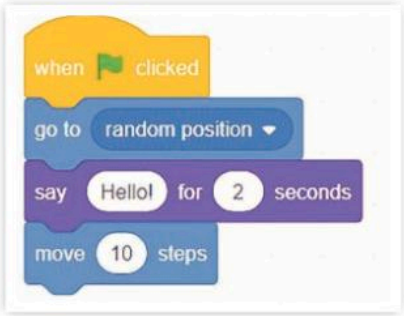
- how to make changes to a program
- how to find and fix errors.

Commands to match actions

A student wanted to make a program. They decided the sprite would:

- jump to a random place
- say "Hello!"
- move around the screen forever.

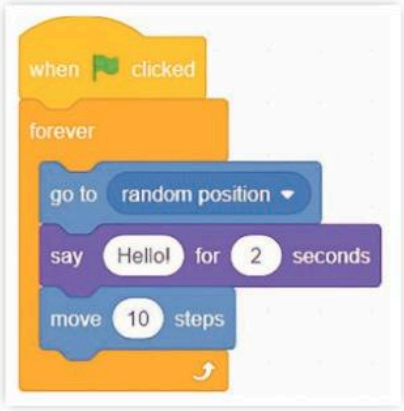
The student found blocks to match the actions. Here is the program made of those blocks.



The student ran the program. The program did not work the way they wanted. The sprite moved just once, then stopped.

Using a loop

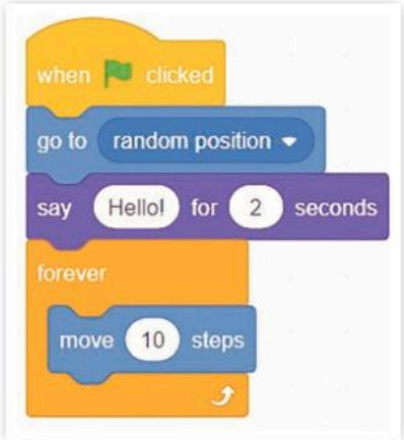
The student decided to use a 'forever loop'. They put all the command blocks inside the loop. Here is the program they made.



The student ran the program. The program did not work the way they wanted. Too many blocks are inside the loop. The sprite keeps jumping to a new place. The sprite keeps saying "Hello!"

Inside and outside the loop

Some commands should go before the loop. Those are the commands that must happen only once.



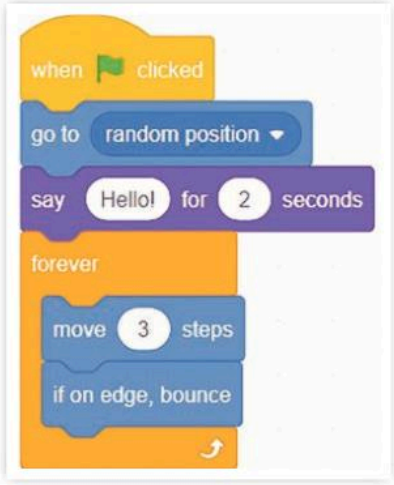
Some commands should be put inside the loop. Those are the commands that will be repeated.

Look at the final picture on the previous page. You will see the program with all the commands in the right place.

Add some bounce

The sprite moves until it is almost off the screen and then stops. You must add a block that tells the sprite to 'bounce' if it hits the side.

The finished program is shown here.





Activity

Using the ideas on this page make a program that makes a butterfly move about the screen.



Extra challenge

Look in the program you made to find the block that says 'move 10 steps'.

Change the number. Run the program and see what is different.

Explore what happens when you use different numbers.



Explore more

Make a new program. Use a sprite that looks like a ball. Make it bounce around the screen.

Add different types of ball. Set them to different speeds.

